

Paper 5*Evaluation of survival data and two new rank order statistics arising in its consideration***Author**

Mantel N

Reference*Cancer Chemotherapy Reports* 1966; 50:163–170**Summary**

The life-table approach developed in 1958 by Kaplan and Meier (Paper 4) was extended to allow comparison of two or more life-tables via the so-called logrank test. Comparison of survival curves was reduced to estimating the common odds ratio in a series of 2×2 tables. The proportional hazards concept was introduced and a simple estimate of the odds ratio was developed as a weighted average of the odds ratios from the individual tables. Sampling properties of the test and estimator were deduced. Extensions to more than two survival curves were also indicated.

Related references (1)

- (1) Mantel N and Haenszel W. Statistical aspects of the analysis of data from retrospective studies of disease. *Journal of the National Cancer Institute* 1959; 22:719–748.
- (2) Mantel N. Chi-square tests with one degree of freedom: Extensions of the Mantel–Haenszel procedure. *Journal of the American Statistical Association* 1963; 58:690–700.
- (3) Peto R and Peto J. Asymptotically efficient rank in variant procedures (with discussion). *Journal of the Royal Statistical Society, Series B* 1972; 135:185–206.
- (4) Prentice RL. Linear rank tests with right censored data. *Biometrika* 1978; 65:167–179.

Key message

An efficient test and estimator for comparing two survival curves was provided that could handle censored data.

Why it's important

This paper provided an efficient and unbiased method for comparing two survival curves that were generated by incomplete data. This test (variably referred to as the Mantel–Haenszel test or the logrank test) has become the standard method for comparing two treatments in a clinical trial. Previous approaches of comparing proportions at a fixed follow-up time or overall crude failure rates, were inefficient and subject to bias in ascertainment or length of follow-up.

Strengths

This is an efficient, robust, stable and intuitive method for comparing two survival curves. In addition to providing a test for the equality of two survival curves, a robust estimator for the relative failure rate of the two arms was given.

Weaknesses

It is not able to handle patient-specific covariates. The estimator of the relative failure rate is inefficient when the differences are large.

Relevance

This remains the standard method for comparing two survival curves when there are no covariates.